COMP3311 Week 1 Monday Lecture

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- Entity-Relationship Data Modelling
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COMP3311 25T1
Database Systems

https://webcms3.cse.unsw.edu.au/COMP3311/25T1/

Lecturer

Name:	John Shepherd
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Consults:	by email, Ure CSE Help!
Rooms:	Convenor, COMP3311 25T1 Deputy Head of School (wha)
Research:	Information Extraction Information Retrieval Text Mining, Text Analytics Multimedia Databases
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What to call me? Jas, John, Dr Shepherd, HRH, Your Majesty

Course Admin

Name:	Dylan Brotherston
Email:	cs3311@cse.unsw.edu.au
Roles:	COMP3311 25T1 Admin Systems Support Extraordinaire

Despite the picture, Dylan is not a bat

Tutors

Many, many students → many tutors ...

- Abbie Worswick
- Alex Lee
- Ayda Orang
- Edward Qian
- Evan Krul
- Hellen Wang
- Henry Li
- Justin Liu
- Kenneth Li
- Matthew Perry
- Neil Dadlich
- Sai Nair
- Soosyung Moon
- Xinzhang Chen

How COMP3311 25T1 will run

Sources of information:

- Topic Videos ... primary content
- Lectures ... summary content + examples
- textbooks ... most detailed version of topics

Activities:

- lectures ... work through examples, ask questions
- tutorials ... work through exercises, ask questions
- prac exercises ... learn the systems and skills
- assignments ... practice your skills
- exam ... demonstrate your knowledge/skills

Lecture Videos

Echo360 records the lecture; I also record the lecture.

After the lecture, Echo360 processing takes ~1.5 hours

After the lecture ...

- I do some post-processing (~1 hour)
- I upload to YouTube (~1 hour)
- I link it to the web site (when I remember)

I do *not* monetize the YouTube videos; any ads are courtesy of Google.

Know your course website

Access via https://webcms3.cse.unsw.edu.au/COMP3311/25T1/

- Home: where Notices and Upcomings appear
- Course Outline: the "contract"
- Timetable: info on tutors and locations for tutorials
- Topic Videos: short-form (20-40 min), topic based videos
- Lectures: slides, videos and exercise solutions
- Assignments: assignment specs and other material
- Tutorials: questions and, eventually, answers
- Prac Work: lab-like practical exercises
- Activities: quizzes and polls

Your DB Server

You run you own PostgreSQL server on the host **`vxdb02`**

How to access the **`vxdb02`** server

- from Vlab: **`ssh vxdb02`**
- from Home: **`ssh YourUserNamevxdb02.cse.unsw.edu.au`**

On the **`vxdb02`** server you have

- your standard CSE directories
- a special directory **`/localstorage/YourUserName/`**

The only software you should run on **`vxdb02`** is PostgreSQL (no vscode)

Exercise: Play with `vxdb02`

Login to **`vxdb02`**

Find your **`/localstorage`** directory

Set up your environment

Install your PostgreSQL server

Explore the files in your PostgreSQL server

More details on doing this are in Prac Exercise 01.

Data Modelling

Aims of data modelling:

- describe what information is contained in the database
  (e.g. entities, students, courses, accounts, branches, patients, ...)
- describe relationships between data items
  (e.g. Lecturer teaches Course, Tom's account is held at Coupes' Gap, John is married to Jane)
- describe constraints on the data
  (e.g. 7 digit IDs, students can enrol in no more than 3 courses per term)

Data modelling is a design process

- converts requirements into a data model

Some Design Ideas

Consider the following while working through exercises:

- start simple ... evolve design as problem better understood
- identify objects (and their properties), then relationships
- represent obvious invariants (keys, foreign keys, ...)
- keywords in requirements suggest data/relationships
  (role of thumb: nouns → data, verbs → relationships)
- don't confuse operations with relationships
  (operation: he buys a book; relationship: the book is owned by him)
- consider all possible data, not just what is available

Exercise: Instagram Data Model

Consider Instagram (photos, videos, viewers, posters, ...)

Develop an informal data model for it by identifying:

- the data items involved (objects and their attributes)
- relationships between these data items
- constraints on the data and relationships

Exercise: GMail Data Model

Consider the GMail system (or any other modern mail client)

Develop an informal data model for it by identifying:

- the data items involved (objects and their attributes)
- relationships between these data items
- constraints on the data and relationships

Exercise: Handbook Data Model

Imagine that we wanted a database for the UNSW Handbook

Work out requirements by looking at the actual Handbook

Develop an informal data model for it by identifying:

- the data items involved (objects and their attributes)
- relationships between these data items
- constraints on the data and relationships

Entity-Relationship Data Modelling

The world is viewed as a collection of inter-related entities.

ER has three major modelling constructs:

- attribute data item describing a property of interest
- entity: collection of attributes describing objects of interest
- relationship: association between entities (objects)

ER diagram: modelling notation containing collections of

- entities (rectangles) and their associated attributes (ovals)
- relationships (diamonds) between entities (1:1, 1:n, n:n, total, partial)
- keys: sets of attributes that uniquely define entities

Lecture notes use notation from SKS and GUW books (simple)

Entity-Relationship Data Modelling (cont)

Example ER diagram:

Entity-Relationship Data Modelling (cont)

ER-design elements:

Entity-Relationship Data Modelling (cont)

Example of attribute notations:

Entity-Relationship Data Modelling (cont)

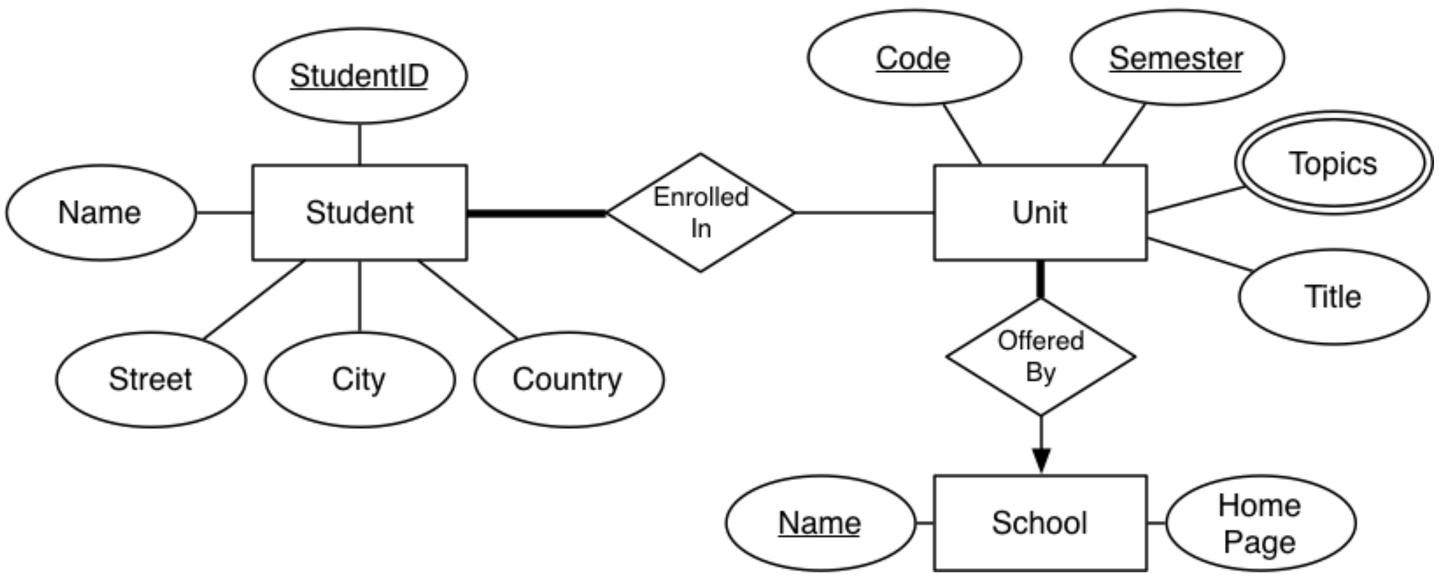
Relationships:

Thick line = total participation; thin line = partial participation

Exercise: Relationship Semantics

Describe precisely the semantics of the following relationships:

❖ Exercise: ER Diagram

Using the ER diagram below

- give examples of entity values
- describe the semantics of the relationships